SYED TAHMID MANZOOR

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EDUCATION

North South University Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering *Expected [December 2026]*
Relevant Coursework: Deep Learning, Machine Learning, Design and Analysis of Algorithms, Data Structures, Database Systems, Operating Systems, Software Engineering, Microprocessors and Interfacing

RESEARCH EXPERIENCE

Longitudinal Brain Tumor Growth Prediction from Multi-Temporal MRI **Year:**2026
Independent Research • github.com/SayeedTahmid/SAILOR_Longitudinal_Research

- Audited a longitudinal MRI prediction method published in IEEE TMI (2025) and showed it was statistically indistinguishable from a copy-forward baseline, tracing this to a conditioning variable acting as a positional proxy and 19 sessions with zero-filled masks.
- Engineered a 33-module Python research package (~4,000 lines, 37 unit tests) with permutation controls and provenance checks to rule out data leakage.
- Preprocessed and quality-checked two glioma cohorts totalling 118 patients (EBRAINS, 27; LUMIERE, 91), identifying reliability failures in LUMIERE automated segmentations.

3D Brain Tumor Segmentation from Pediatric MRI **Year:**2026
Senior Design Project I (CSE499A) • Advisor: Riasat Khan, Associate Professor, ECE

- Lead the deep learning pipeline for volumetric tumor segmentation on BraTS-PEDs 2024 (257 labeled cases) in a three-person team.
- Rescoped the problem, dataset, and evaluation protocol under a fixed deadline after the initial approach proved unviable.

Multimodal Gas Detection: Reproduction, Critique, and Hybrid Model **Year:**2026
Deep Learning (CSE465)

- Reproduced a 2025 published gas-detection study and uncovered 602 duplicate images, a 300× undocumented resolution discrepancy, and two reportedly distinct models with identical predictions (McNemar p = 1.0).
- Showed a depth-5 decision tree matched the published deep model under random splits but collapsed under temporal splits, isolating an evaluation-protocol artifact.
- Designed a hybrid ViT + 1D-CNN architecture (ViT-B/16 with LoRA, FT-Transformer) evaluated with ablation, calibration, and open-set recognition.

PROJECTS

Customer Relationship Management System Junior Design Project (CSE299) Year:2025
Python/Flask, React, TypeScript, Firebase, Kotlin • github.com/SayeedTahmid/CRMS_Junior_Project_CSE299

- Built a multi-tenant CRM spanning a Flask REST API, React/TypeScript web client, and native Kotlin Android app, with role-based access control and SLA-tracked complaint handling on Firestore.
- Integrated Gmail API email tracking via n8n workflows, a Telegram notification bot, and VoIP call management into one customer timeline.

OS Algorithm Simulator & Performance Analyzer os-algo-sim-plum.vercel.app **Year:**2026
React 18, TypeScript, Vite, Tailwind CSS, Recharts • github.com/SayeedTahmid/OS-AlgoSim

- Implemented 12 OS algorithms client-side with no backend: 6 CPU schedulers, 4 memory-management algorithms, and Banker's algorithm with deadlock detection.
- Built a comparative analyzer benchmarking schedulers on a shared workload across waiting, turnaround, and response time, with Gantt and time-series charts.

Smart Flood Early Warning System Microprocessors & Interfacing (CSE331L) **Year:**2026
Bare-metal C, STM32F103C8T6 (ARM Cortex-M3), STM32CubeIDE

- Fused ultrasonic, humidity, and soil-moisture inputs into a weighted flood risk index driving a three-tier alert state machine with hysteresis, on a bill of materials under BDT 2,000.
- Wrote interrupt-driven firmware with the HAL disabled for direct register access: a 1 ms SysTick ISR computes water rise rate without blocking a 100 ms superloop driving an I2C OLED, LED bank, and PWM buzzer.

Crowd Action Detection Vision-Language Model **Year:**2026
PyTorch, OpenAI CLIP (ViT-B/32) • github.com/SayeedTahmid/Crowd-action-detection-VLM

- Reached 96% test accuracy across three crowd action classes by adapting CLIP as a frozen vision-language backbone with a custom classifier head; deployed live on Hugging Face Spaces.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, Kotlin, SQL
Machine Learning: PyTorch, Vision Transformers (ViT, CLIP), LoRA fine-tuning, 3D medical image segmentation, model calibration, ablation design
Web & Backend: Flask, REST APIs, React, Tailwind CSS, Vite, Firebase (Firestore, Auth), n8n
Embedded & Tools: Bare-metal C on STM32 / ARM Cortex-M, interrupts, I2C, ADC, PWM; Git, Google Colab, Hugging Face, Vercel, LaTeX, pytest, CUDA

EXTRACURRICULAR

Football, North South University main team — ASCENT Cup and BGMEA corporate tournaments. Member, NSU Games and Sports Club.